

TUS OS — Topology-Verified Decision Receipts (Concept Note)

v1 (public concept note) — 2026-02-14

Protocol-style overview: architecture, artifact taxonomy, terminology, and minimal example

Abstract

TUS OS is a protocol for producing cryptographically verifiable decision receipts. A receipt is a canonical, hash-addressable record of a decision engine's structural behavior (state, candidate moves, ranking, selection, and optional lookahead) that can be independently verified later. TUS OS separates evidence (receipts) from interpretation (bindings and value overlays), enabling policy plurality without fragmenting the underlying evidence.

Scope and non-claims

This note is a conceptual overview. It does not publish the full normative specification, deterministic numeric contract, canonicalization rules, or conformance suite. References to GUIDANCE and FORECAST are descriptive only. In particular, FORECAST is non-prescriptive and must not be interpreted as prediction, recommendation, or advice.

Model (high level)

TUS OS operates over a finite binary state space. In the reference presentation used throughout this note, the state is a 6-bit vector; the induced topology is the 6-dimensional hypercube (each move flips exactly one bit / line). The same receipt pattern generalizes to other fixed dimensions when declared.

Design goals

- 1) Verifiable evidence: decisions can be checked for integrity and internal consistency.
- 2) Separation of concerns: kernel evidence is independent of domain semantics and value judgments.
- 3) Deterministic auditability: receipts are designed to support reproducible verification.
- 4) Composable packaging: bundles can be sealed with manifests and referenced by digest.

Terminology (minimal)

- Line: a 1-indexed bit position in the state vector.
- State: a D-bit vector, typically encoded as hex for transport.
- Neighbor: a state reachable by flipping exactly one line.
- Canonical receipt: the authoritative machine-verifiable JSON record.
- Audit view: a human-readable rendering derived from a canonical receipt.

Artifact taxonomy

- Canonical receipt (JSON): the authoritative machine-verifiable record.
- Audit view (rendered): a human-readable view derived from a canonical receipt.
- Policy pack (binding): a hash-addressed bundle of external semantics / overlays (optional).
- Manifest (bundle): a hashed index of files and digests for a decision bundle.

Operational modes (high level)

- GUIDANCE: deterministic selection under declared rules (e.g., argmax over line scores).
- FORECAST: structured exploration of alternatives under declared rules; non-prescriptive by design.

Minimal example (illustrative)

State space: D=6 binary lines; state encoded as hex.

Canonical receipt (excerpt):

{

```
"mode": "GUIDANCE",
"from_state_hex": "28",
"p_line": [0.0685, 0.1117, 0.0181, 0.5018, 0.1284, 0.1716],
"chosen_line": 4,
"flip_mask_hex": "08",
"to_state_hex": "20",
"receipt_hash_sha256": "...
}
```

Audit view (excerpt):

Ranked candidate lines by p_line: 4 > 6 > 5 > 2 > 1 > 3. Selected line: 4. Lookahead: present.

Status

A normative specification, reference verifier, and conformance materials will be published separately.

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Author: Mark Whitlock

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